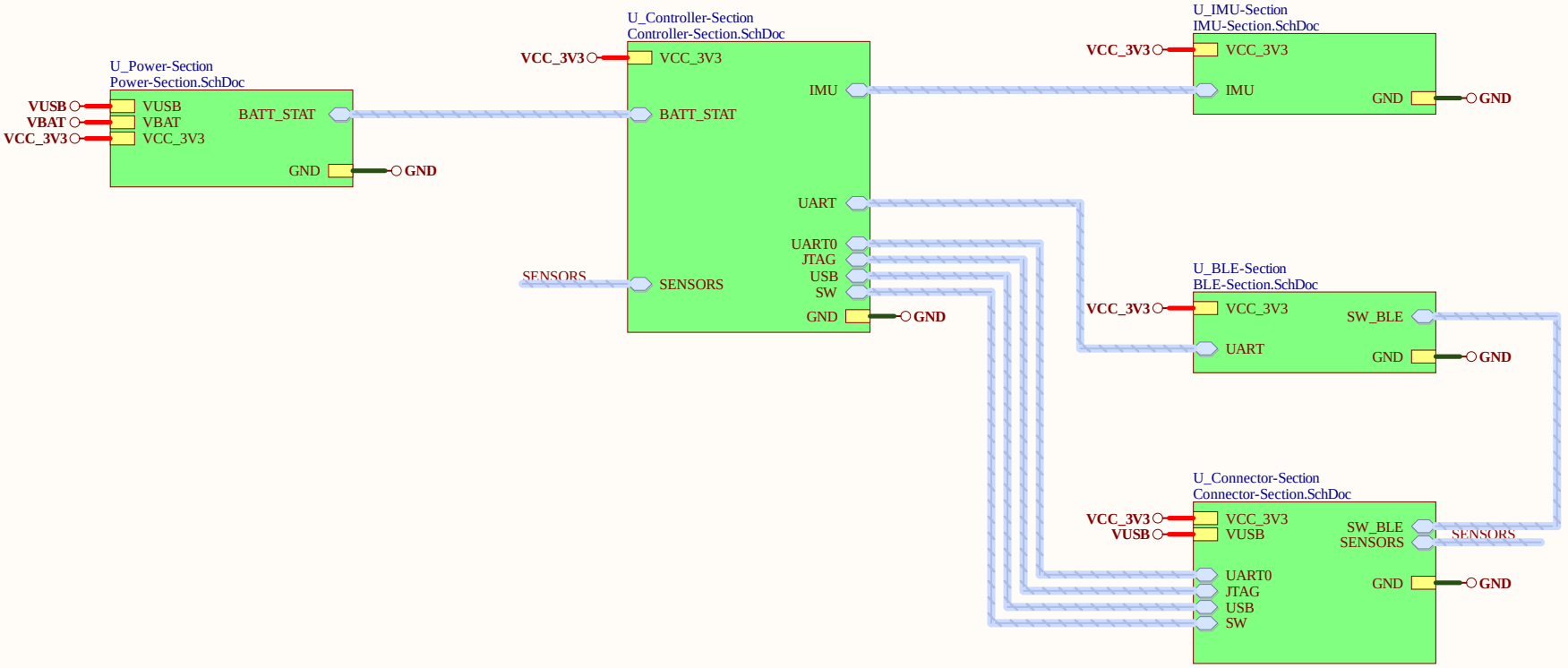
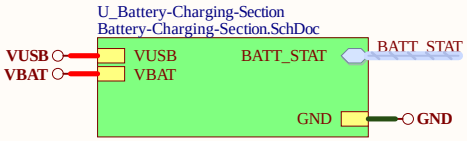
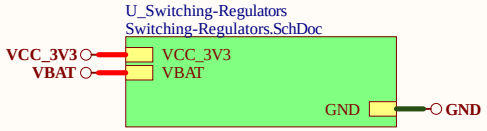
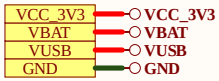


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: ArmBand-Control-Board.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 2/08/2025	Sheet 1 of 8	
	File: S:\ManarHarake-sEMGDevice\..\Top-Sheet.SchDoc	Drawn By: Huzaifa Ali	









Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.

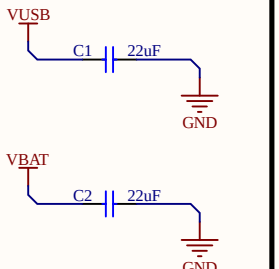
Title	ArmBand-Control-Board.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	2/08/2025	Sheet 2 of	8
File:	S:\ManarHarake-sEMGDevice\..\Power-Section.SchDoc	Drawn By:	Huzaifa Ali

BATTERY STATUS

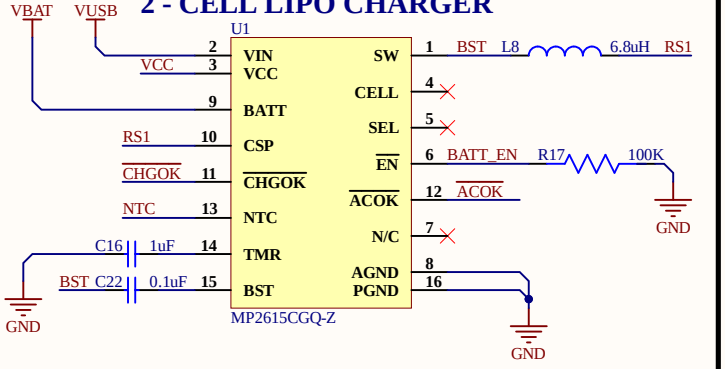


CELL is floating for 2 cells
SEL pin is Floating for 4.2/cell terminating voltage
IC=52.5mV/Rs1 mOhm
IC=2.1A
NTC sensing not used.

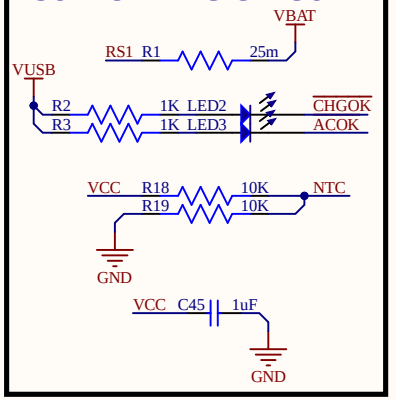
DECOUPLING CAPS

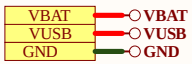


2 - CELL LIPO CHARGER



SUPPORTING CIRCUIT

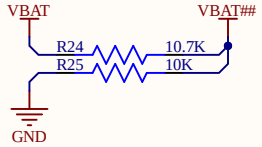




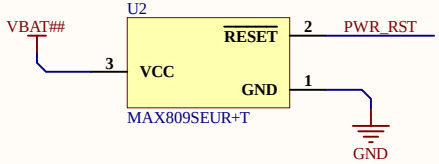
Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.

Title: ArmBand-Control-Board.PrjPcb		
Size: A4	Number: 1	Revision: 1
Date: 2/08/2025	Sheet 3 of 8	
File: S:\ManarHarake-sEMGDevice\..\Battery-Charging\Section_SchDoc	Drawn By: Huzaifa Ali	

VDR FOR 6V to 2.9V

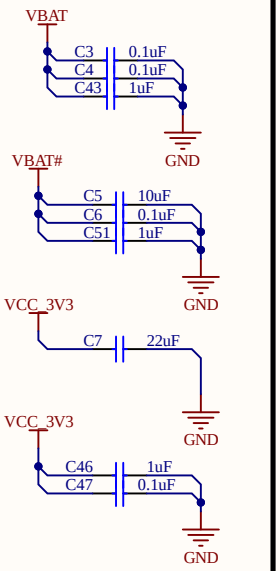


2.9V BATTERY SUPERVISOR

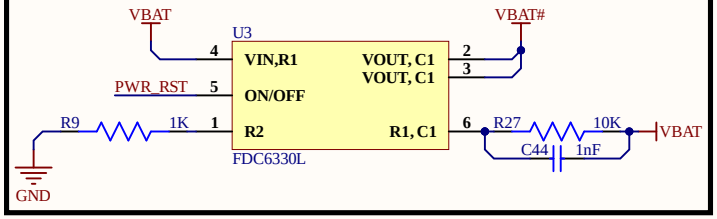


RESET Output remains low while VCC is below the reset threshold, and for at least 140ms after VCC rises above the reset threshold.
Datasheet pg 5

DECOUPLING CAPS

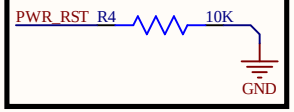


HIGH-SIDE SWITCH

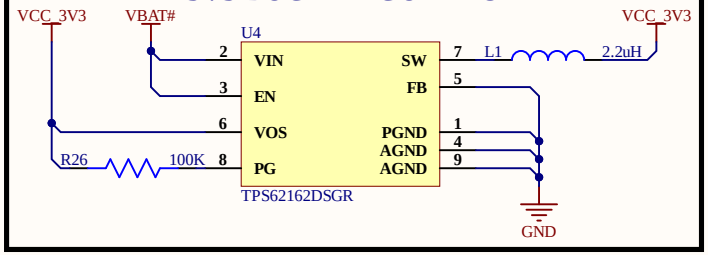


U3 will turn off when ON/OFF is low.
U3 will be enabled when ON/OFF is high.
VON/OFF= 1.5V-8V
VIN= 3-20V

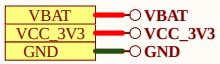
R-PULL DOWN



3V3 BUCK REGULATOR



Fixed regulator.
VOUT= 3V3/1A

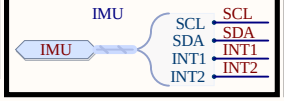


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: ArmBand-Control-Board.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 2/08/2025	Sheet 4 of 8	
	File: S:\ManarHarake-sEMGDevice\..\Switching Regulators.SchDoc	Drawn By: Hamza Ali	

BATTERY STATUS



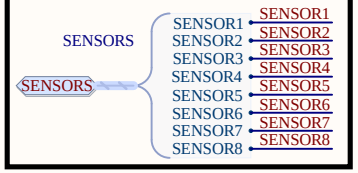
IMU SIGNALS



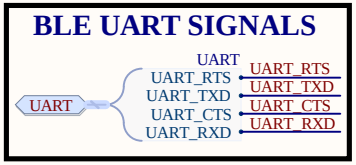
USB SIGNALS



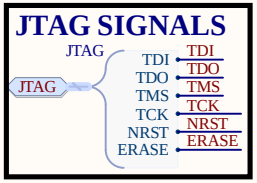
SESNOR SIGNALS



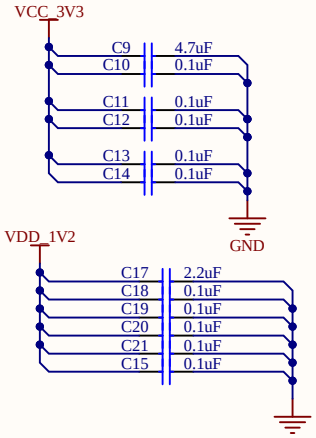
BLE UART SIGNALS



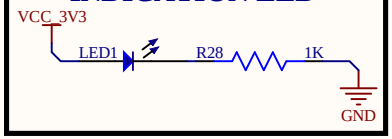
JTAG SIGNALS



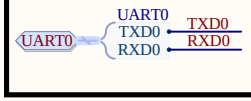
DECOUPLING CAPS

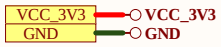


INDICATION LED

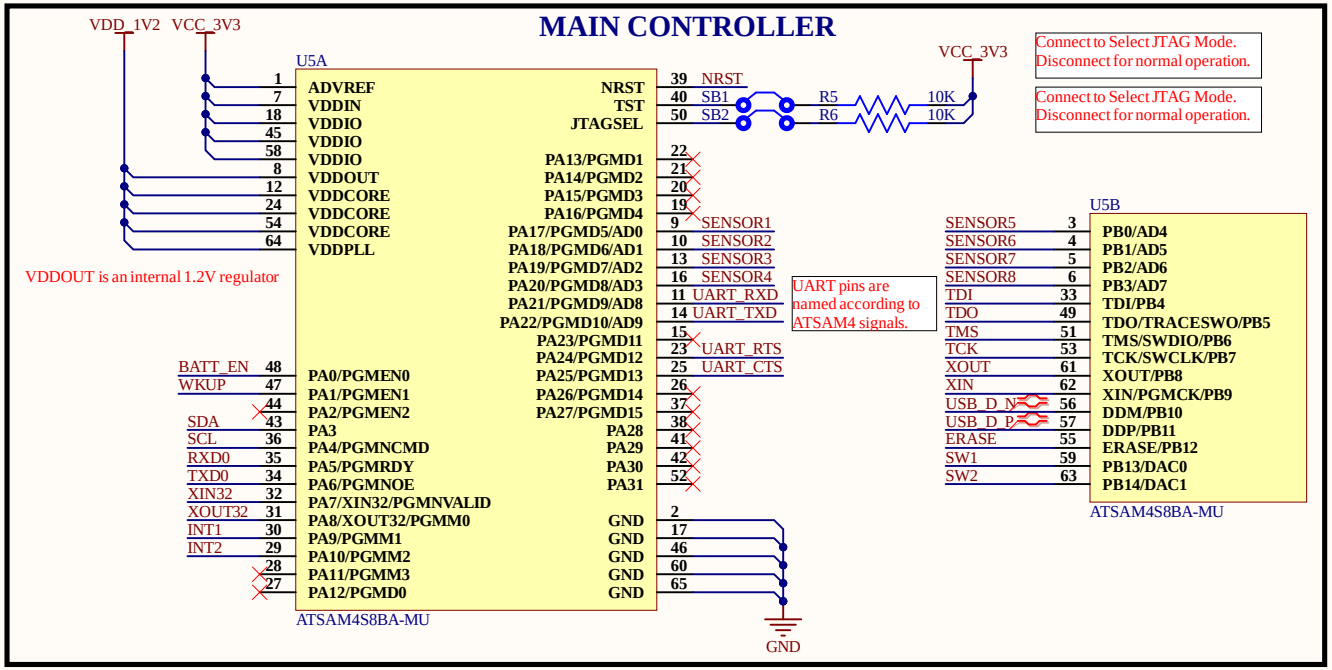


UART SIGNALS

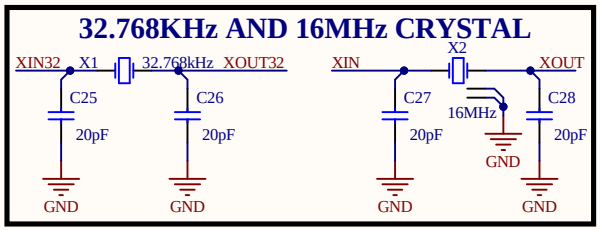




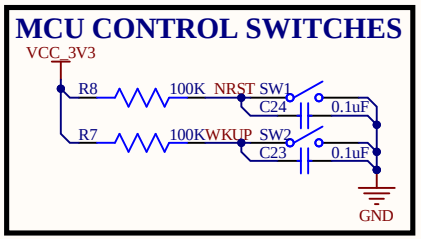
MAIN CONTROLLER



32.768KHz AND 16MHz CRYSTAL



MCU CONTROL SWITCHES



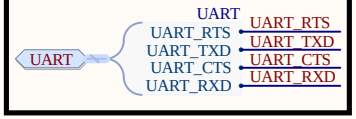
SWITCH SIGNALS



Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.

Title: ArmBand-Control-Board.PrjPcb		
Size: A4	Number: 1	Revision: 1
Date: 2/08/2025	Sheet5 of 8	
File: S:\ManarHarake-sEMGDevice\..\Controller\Sec...SchDoc	Drawn By: Hozaifa Ali	

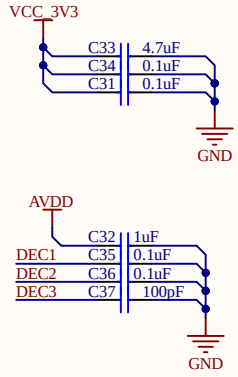
BLE UART SIGNALS



SW SIGNALS

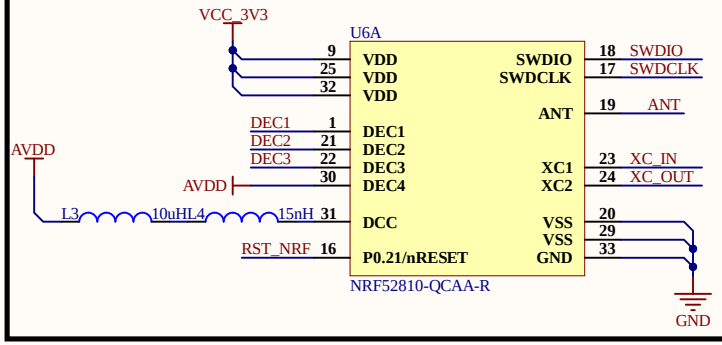


DECOUPLING CAP

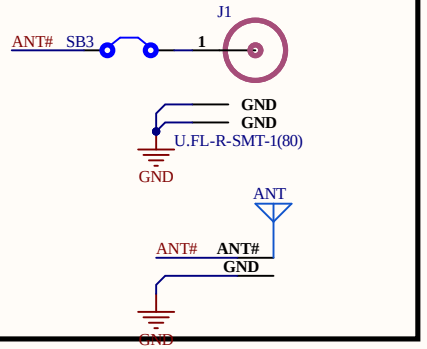


C36 may be disconnected if not needed

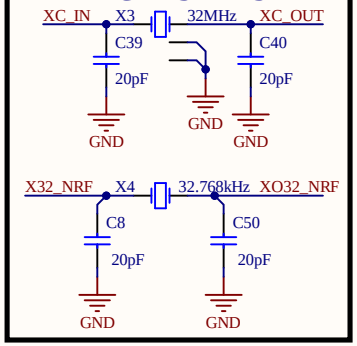
BLE TRANCIEVER POWER BANK



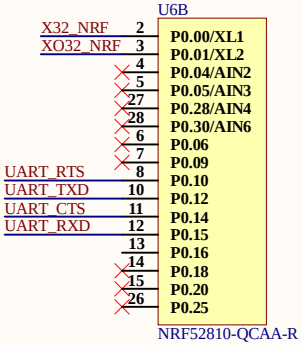
ANTENNA CONN



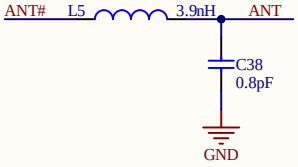
CRYSTALS

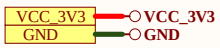


BLE TRANCIEVER I/O BANK



ANTENNA SUPPORTING CIRCUIT

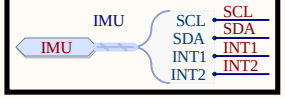




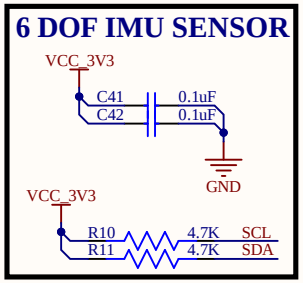
Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.

Title	ArmBand-Control-Board.PrjPcb	
Size: A4	Number: 1	Revision: 1
Date: 2/08/2025	Sheet 6 of 8	
File: S:\ManarHarake-sEMGDevice\..\BLE-Section.SchDoc	Drawn By: Huzaifa Ali	

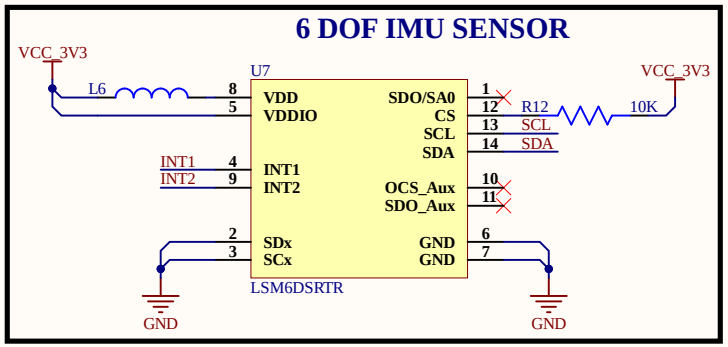
IMU SIGNALS



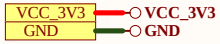
6 DOF IMU SENSOR



6 DOF IMU SENSOR

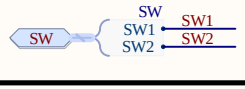


CS logic 1 to enable I2C and disable SPI mode.

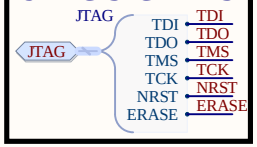


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: ArmBand-Control-Board.PrjPcb	
	Size: A4 / Number: 1	Revision: 1
	Date: 2/08/2025 / Sheet 7 of 8	
	File: S:\ManarHarake-sEMGDevice\..\IMU-Sensor.SchDoc	Drawn By: Huzaifa Ali

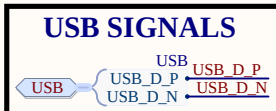
SWITCH SIGNALS



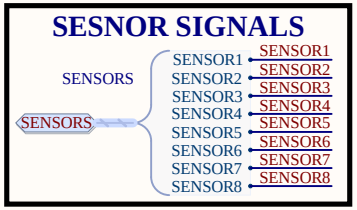
JTAG SIGNALS



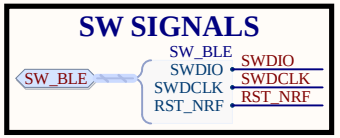
USB SIGNALS



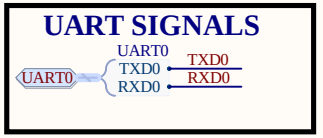
SESNOR SIGNALS

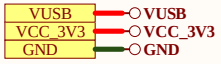


SW SIGNALS

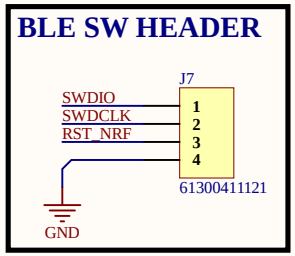


UART SIGNALS

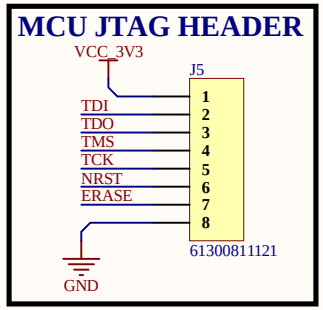




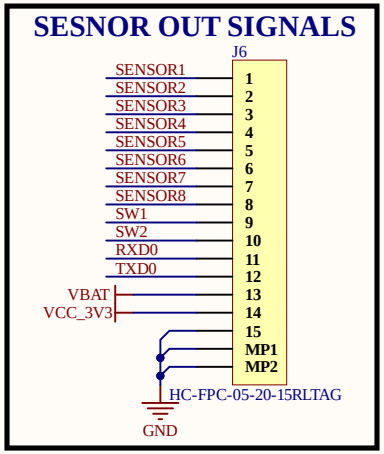
BLE SW HEADER



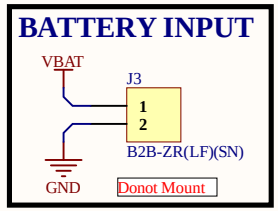
MCU JTAG HEADER



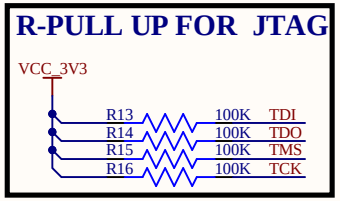
SESNOR OUT SIGNALS



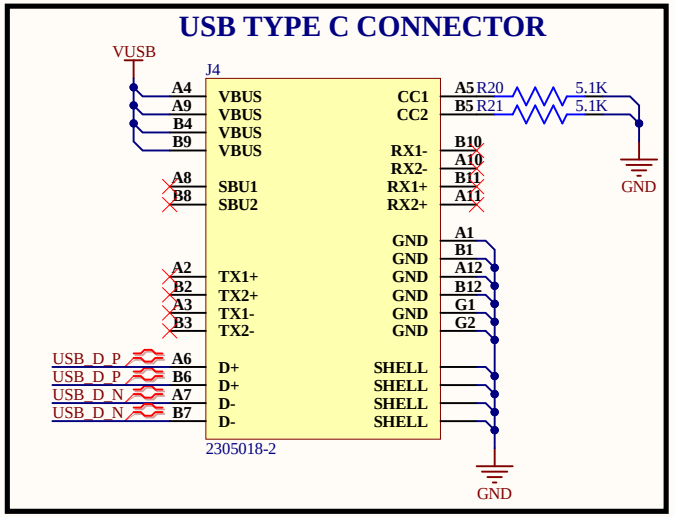
BATTERY INPUT



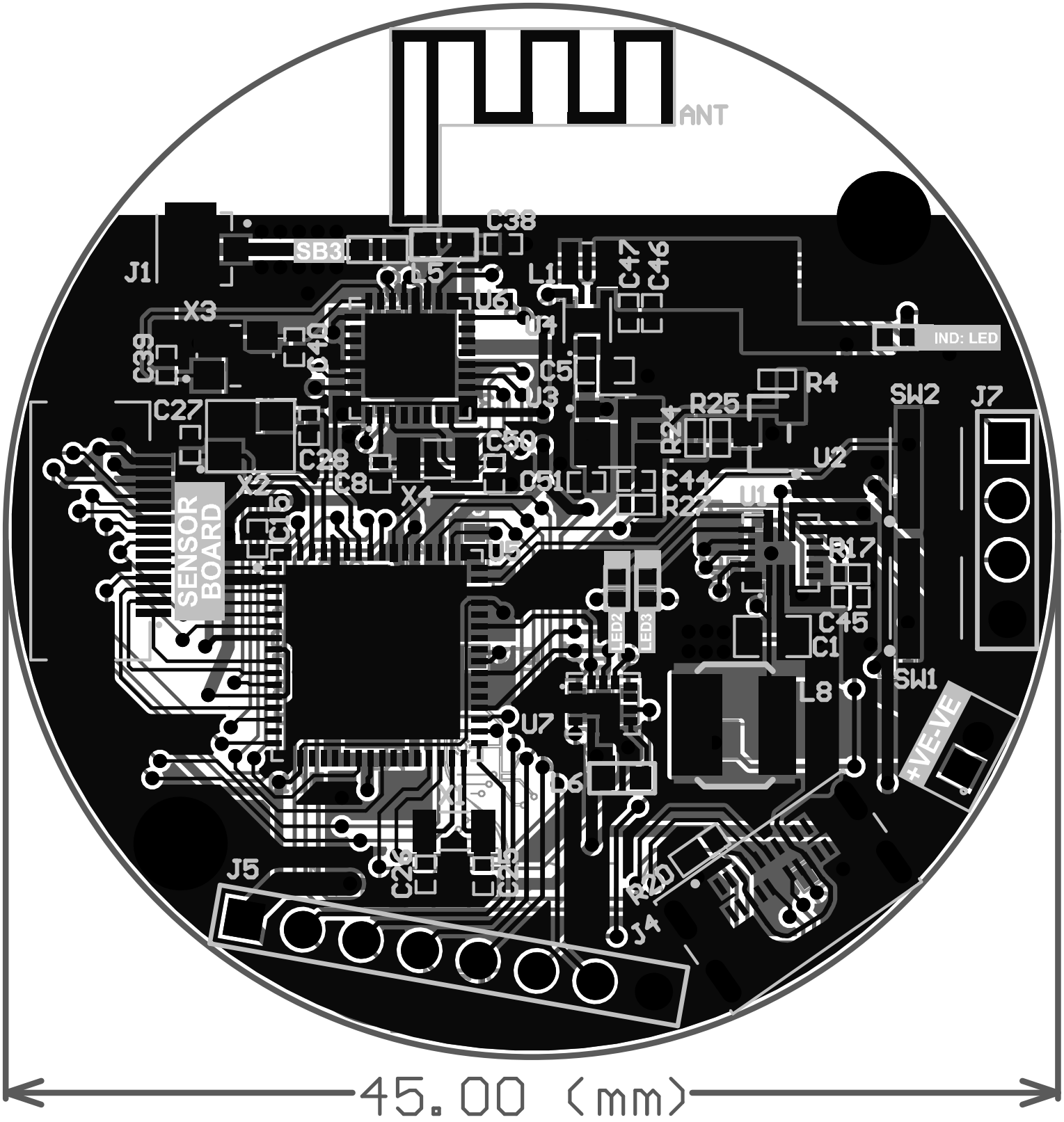
R-PULL UP FOR JTAG



USB TYPE C CONNECTOR



Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: ArmBand-Control-Board.PrjPcb	
	Size: A4 / Number: 1	Revision: 1
	Date: 2/08/2025 / Sheet 8 of 8	
	File: S:\ManarHarake-sEMGDevice\..\Connectors.SchDoc	Drawn By: Hozaifa Ali


ANT
J1
SB3
C38
L1
C47
C46
X3
U6
U4
C39
C49
C5
U3
IND: LED
R4
R25
SW2
J7
C27
C50
R24
C28
C8
X2
X4
C51
C41
R23
U1
U2
C16
SENSOR BOARD
U5
R17
LED2
LED3
C45
C1
L8
SW1
+VE-VE
U7
D6
X1
C20
C21
R10
J5
J4
45.00 (mm)